

Mining Multi-Scale Spatial-Frequency Clues for Unsupervised Intrusion Detection

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Abstract—Unsupervised network-based intrusion detection system (UNIDS) identifies suspicious traffic and alerts administrators without using any traffic labels. Existing Graph Convolutional Network (GCN)-based UNIDS approaches show great potential with collaboratively utilizing traffic features and network topologies. However, these methods suffer from an excessively high false-positive rate (FPR), e.g., 3.27% FPR in a supervised NIDS approach, while increases dramatically to 19.37% under the unsupervised setting. We reveal that the high FPR stems from a single-scale spatial-frequency learning paradigm, which blurs the distinction between benign and malicious traffic and misleads UNIDS systems. Therefore, we propose a Multi-scale Spatial-Frequency Intrusion Detection System (MSF-IDS) to mitigate the high FPR. Specifically, we propose multi-scale frequency encoders, thereby differentiating abnormal feature patterns. Then we propose NAPH as a spatial encoder, mining intrinsic abnormal topology patterns by tracking tens of thousands of evolving traffic nodes. To the best of our knowledge, NAPH takes the first step toward differentiable persistent homology analysis over dynamic network data. We also develop an executable application for NAPH to provide easy-access visualization insights. Finally, a self-supervised representation augmentor and an intrusion detector are proposed to refine and highlight the attack-specific information. Equipped with MSF-IDS, administrators effectively identify the unknown attack traffic, freeing security staff from labor-intensive engineering. Extensive experiments demonstrate the superiority of MSF-IDS, including binary classification, multi-classification, online intrusion detection, and visualized discussions. Our codes,

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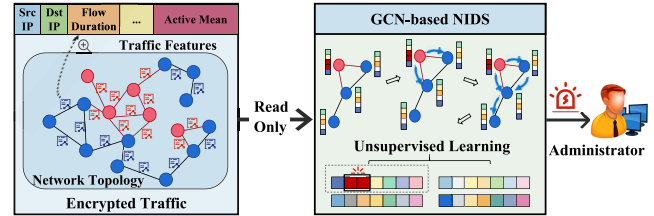


Fig. 1. Illustration of a GCN-based UNIDS. The left is encrypted traffic data, with only available traffic features and network topology. The former includes the statistical description of the traffic flow, i.e. source IP, flow duration, etc. The network topology and features are fed into UNIDS in a read-only way. Then the UNIDS aggregates the traffic features by topology in an unsupervised way and alerts the suspected ones to administrator for further response. More specific threat model of this paper is available in Appendix P (see supplementary material).

datasets, and an executable application are available at <https://github.com/qcydm/MSF-IDS>

Index Terms—Unsupervised intrusion detection, network security, anomaly detection.

I. INTRODUCTION

NETWORK intrusion detection system (NIDS) [1], [2], [3], [4] forms a security frontline in network systems, aiming to filter out malicious network attacks in an automatic way, such as man-in-the-middle attacks (MITM) [5], and denial of service (DoS) [6]. These attacks can be considered outliers or anomalies, as they may involve different characteristics in **both traffic features and traffic topologies** from the benign ones [7]. However, most of the existing NIDS methods are built on inherent anomaly patterns, e.g. blacklist [8], whitelist [9], or known attack labels [10], struggling to identify new attacks accurately, such as sophisticated phishing schemes, using deep fake technology to impersonate trusted sources [11]. With the rapid development of mobile online business and wireless communications [12], [13], [14], [15], unknown network attacks show an escalation in both sophistication and frequency. According to Splunk's "State of World Security 2023" [16], 65% of respondents indicated that they have observed an increase in attempts at unknown attacks, a 39% increase over the previous year's survey. The surge of unknown attacks in mobile online businesses leaves security teams exhausted and overextended. This urgently calls for an intelligent unsupervised network-based intrusion detection

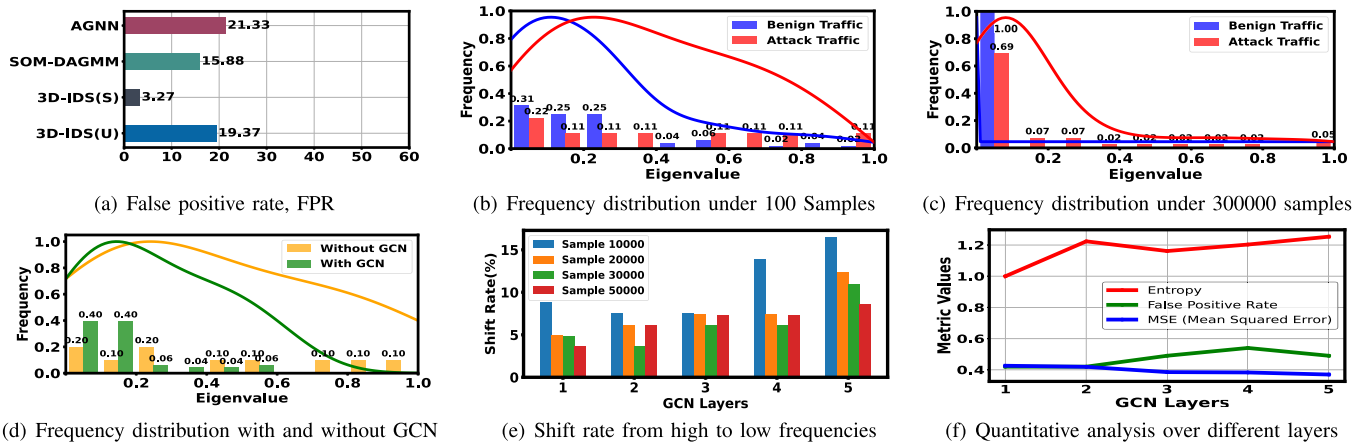


Fig. 2. (a)(b)(c)(d) show that the GCN-based UNIDS filters out high-frequency information in spectral view, referred to as single-scale frequency learning, and leads to high FPR. (e)(f) show that the single-scale topology learning exacerbates FPR in spatial view.

system (UNIDS) capable of identifying unknown and evolving attacks, especially for the abnormal spatial-frequency patterns.

The existing unsupervised NIDS methods involve three categories: the signature-based ones [17], the anomaly-based ones [18], [19], and the deep learning-based ones [20], [21]. The signature-based ones are built on pre-defined anomaly patterns, i.e. blacklist [22]. The attack traffic can easily escape, once the pre-defined patterns are not matched. The anomaly-based ones can automatically mine the attack patterns in a data-driven way, i.e. clustering methods [23], [24], one-class classifications [25], [26], etc. Lately, deep learning-based ones are popular in UNIDS, freeing security staff from labor-intensive hand-crafted traffic feature engineering, i.e. self-supervised [27], [28], [29], [30], reconstruction-based [27], [31], [32], and graph convolution network-based [33], [34], [35], [36], [37], [38]. The GCN-based UNIDS utilizes both traffic features and topology for intrusion detection, as shown in Fig 1, showcasing great potential in UNIDS tasks.

Although promising, our empirical analyses indicate that existing GCN-based UNIDS faces a high false positive rate (FPR) problem. As shown in Fig. 2(a), we conduct unsupervised intrusion detection on three representative baselines and compare the FPR on the CIC-TON-IoT dataset [39]. AGNN [40], a GCN-based approach, achieves 88.48% F1-score, outperforming the deep-learning baseline SOM-DAGMM [21] at 78.13%. However, AGNN's FPR is 21.33%, significantly higher than the 15.88% of SOM-DAGMM. Furthermore, we report the FPR of an existing SOTA-supervised intrusion detection approach 3D-IDS(S) [41], as low as 3.27%. Then we utilize a cluster model to replace the original classifier for unsupervised intrusion detection in 3D-IDS, referred to as 3D-IDS(U). Under unsupervised settings, the FPR of 3D-IDS(S) rapidly increases to 19.37%, almost 5 times than 3D-IDS(S). These results pinpoint the high FPR as the key bottleneck of GCN-based UNIDS. **Boosting GCN-based UNIDS performance by mitigating its FPR rates is feasible and prospective.**

We attribute the high FPR problem to **single-scale spatial-frequency learning** in GCN-based UNIDS. We present discussions Q1 and Q2 to understand why this scheme makes

traffic representations indistinguishable and misleads UNIDS. In short, single-scale spatial-frequency learning focuses only on a narrow frequency range and filters out high-frequency attack information, causing traffic representations blur between benign and malicious traffic, thus leading to a high FPR.

Q1: Single-scale frequency learning. From the spectral view, GCN-based UNIDS are **low-pass filters** [18], [42]. They collaboratively capture feature and topology information by aggregating neighboring traffic features [43]. In this way, potential attack information will be filtered out, making it hard for UNIDS to differentiate between benign and attack traffic, thus causing a high false positive rate. As shown in Fig. 2(b) (c), we depict the frequency distribution of benign and attack traffic in 100 and 300,000 samples, respectively.¹ We observe that the frequency distributions of attack traffic have significant 'right-shift' phenomena, compared to the benign traffic. It reveals that high-frequency information commonly exists in attack traffic, an important evidence for distinction. Fig. 2(d) further shows that GCN-processed features undergo a left-shift, compared to the without one. This observation uncovers that the GCN learning paradigm will **filter out** and suppress the high-frequency information in attack traffic.

Q2: Single-scale spatial learning. Moreover, the topology learning of GCN-based UNIDS exacerbates the suppression of important high-frequency traffic information. As shown in Fig. 2(e), 'left-shift' phenomena exist in different layers of GCN, and the shift rates increase as the GCN layer increases. Finally, we verify our hypothesis quantitatively using three metrics: entropy, measuring the redundancy or uncertainty, FPR, and mean squared error, measuring the representation similarity. As shown in Fig. 2(f), the entropy and FPR increase with the GCN layer, aligning with the hypothesis on single-scale spatial learning, while MSE decreases, revealing that traffic representations become increasingly similar and indistinguishable between attack and benign traffic.

In response to the above discussion, we raise a crucial question: 'How can a UNIDS model **automatically preserve**

¹More frequency distribution analyses are available in the Appendix L (see supplementary material).

TABLE I

THE COMPARISONS OF THE RELATED WORKS, WHERE ✓ MEANS THAT HAS THE CAPABILITY, AND × MEANS THAT DOES NOT HAVE THE CAPABILITY

	Works	Multi-scale Spatial	Multi-scale Frequency	Multi-scale Spatial-Frequency	Self-supervised Refinement
GCN-based UNIDS	EULER [33]	✓	×	×	×
	AGNN [40]	✓	×	×	✓
	APPNP [63]	×	✓	×	×
	ClenshawGCN [64]	×	✓	×	✓
	MSF-IDS (Ours)	✓	✓	✓	✓
	Works	Dynamic Data	End-to-End Approach	Scalability	Developed Application
Multi-scale Topological Method	PHs [47]	×	×	×	×
	PH for Networks [48]	×	×	×	×
	PIs [65]	×	✓	×	×
	WKPI [66]	×	✓	×	×
	NAPH (Ours module)	✓	✓	✓	✓

multi-scale spatial-frequency traffic information to high-light attacks, thus reducing FPR and boosting performance?”. Achieving this goal is challenging. First, existing spectral GCNs focus on multi-scale frequency preserving, while the effective multi-scale topology learning is underexplored [43], [44], [45], [46]. Second, the persistent homology (PH) approach [47], [48] showcases great potential in multi-scale spatial learning in anomaly detection [49], [50], biomedicine [51], and finance [52], etc. However, PH faces two challenges when deployed in real-world dynamic intrusion detection:

- **Challenge 1. Static data only:** PH is specially designed for static data, e.g., data cloud. Since traffic data evolve over time, efficiently mining abnormal spatial patterns from dynamic traffic streams remains challenging.
- **Challenge 2. Scalability:** PH provides multi-scale spatial insights through persistent diagrams, but the scalability of this paradigm is insufficient for large-scale dynamic traffic data. Bridging the gap between PH and an end-to-end UNIDS framework for tens of thousands of evolving traffic nodes remains challenging.

We propose a novel UNIDS approach MSF-IDS, aiming to mitigate the high FPR and boost detection performance. It has three key components, multi-scale spatial-frequency encoders, a self-supervised representation augmentor, and an unsupervised intrusion detector. In the proposed spatial-frequency encoders, we present wavelet-based dual graph filters to efficiently **preserve multi-scale frequency information**. As for spatial encoders, we propose a **differentiable** network-aware persistent homology, i.e., NAPH, to track the abnormal multi-scale spatial patterns. Specifically, we propose a time filtration for spatial mining over dynamic networks and address Challenge 1. Then we propose a novel persistent heatmap tool for end-to-end intrusion detection, benefiting abnormal topological mining in tens of thousands involving traffic nodes, and address Challenge 2. Finally, the augmentor and detector refine and highlight attack-specific information without using any prior knowledge. Our specific contributions are as follows:

- We present MSF-IDS, a novel unsupervised intrusion detection approach to mitigate the false positive rate. To the best of our knowledge, we are the first to qualitatively and quantitatively uncover the underlying reason.

- We propose multi-scale spatial-frequency encoders to differentiate the salient anomaly patterns in traffic topologies and features, boosting the performances of unsupervised intrusion detection.
- We firstly propose a novel NAPH module to track multi-scale topological information over dynamic networks. NAPH can be seamlessly incorporated into end-to-end deep learning approaches. We also developed an easy-access NAPH app for visualized topological insights.
- We demonstrate the effectiveness of the proposed MSF-IDS in various benchmarks, including binary classification, multi-classification, online intrusion detection, visualized discussions, etc.

II. RELATED WORK

A. Unsupervised Network Intrusion Detection

The existing unsupervised network intrusion detection systems (UNIDS) can be classified into three categories: the signature-based ones [17], the anomaly-based ones [18], [19], and the deep learning-based ones [20], [21]. In the deep learning-based ones, the state-of-the-art methods are the GCN-based ones [33], [34], [35], [37]. Zhang et al. [53] systematically reviews recent GCN-based intrusion detection approaches, pointing that ‘reducing the FPR within a secure range’, which poses practical challenge to existing GCN-based methods and also aligned with our study motivation. Among them, GLD-Net [54] augments traffic features with dynamic DDoS topology graphs, but still relies on vanilla GCN layers that suffer from over-smoothing. GLAD-PAW [55] leverages weighted graph attention to capture session-level anomalies, yet its graph-level pooling strategy aggregates all nodes uniformly, thereby diluting fine-grained multi-scale patterns. Most related to our works is EULER [33], an unsupervised GCNs and Recurrent Neural Networks (RNNs) stacking-based dynamic intrusion detection technique. The key differences between our MSF-IDS and EULER lie in the following two aspects: 1) MSF-IDS preserves multi-scale feature information to mitigate high FPR, while EULER tends to filter out the high-frequency anomaly information by using traditional GCN schemes. 2) MSF-IDS mines the anomaly topology patterns by tracking multi-scale topology information, while EULER uses a single-scale spatial learning strategy, making representations blur and misleading the UNIDS.

B. Spectral Graph Neural Networks

Frequency-based graph neural networks, or graph filters, focus on handling the GCNs by graph signal optimization [56], [57], [58], [59], [60]. The pioneering work is the GCN [43], a low-pass filter, using a simplified first-order Chebyshev kernel. The following works focus on different-pass filters via various optimizations, i.e. low-pass, high-pass, and all-pass. Additionally, there is a line of works designing graph neural networks in spectra view, e.g. ESGNN [61], BWGNN [62]. The frequency-based GCNs have shown a broad picture and potential in graph mining e.g. network intrusion detection. BWGNN is the most similar work to us. Ours differs from it in two aspects: 1) We utilize dual graph filters to preserve the multi-scale traffic frequency information more efficiently, while BWGNN uses a single graph filter to preserve information; 2) We propose a persistent homology-based approach to mine traffic networks explicitly, while BWGNN has no other explicit topology learning design.

C. Persistent Homology and Multi-Scale Topology Mining

Persistent Homology [47], [48], [67], [68] analyzes the topological structure across different scales. As a Topological Data Analysis (TDA) approach, PH identifies and tracks the ‘birth-death’ progression [68], such as connected components, holes, and voids in specific topologies. Specifically, the ‘birth’ of a homology class is defined as the scale ϵ_i at which it first appears, while ‘death’ refers to the smallest subsequent scale ϵ_j where it no longer exists [47]. In this way, dynamic changes at different scales can be used for downstream tasks, such as intrusion detection. More background about PH are available in Appendix J (see supplementary material). Persistent homology has been applied in various fields, including biology [51], finance [52], and anomaly detection [49], [50]. However, to the best of our knowledge, the existing PH approaches are established on static data points or data point clouds [68]. Second, scalability remains an ongoing challenge in this field, especially when dealing with large-scale dynamic datasets [65]. To address these two challenges, we propose NAPH, a differentiable PH approach, achieving end-to-end multi-scale topological analysis over large-scale dynamic network.

In multi-scale topological information mining, a line of UNIDS focused on multi-scale topology in traffic [21], [36], [69], [70], [71], [72]. Most related to our work is DMTR [36]. The key differences between DMTR and our work are two-fold: first, DMTR preserves only multi-scale topology information; second, we propose an augmentor and an intrusion detector to refine and highlight attack-specific information, while DMTR directly utilizes topology to identify anomalous traffic.

Finally, we provide a comparison Table I. Specifically, compared to the existing methods, the proposed MSF-IDS achieves multi-scale spatial-frequency information mining and self-supervised refinement to effectively reduce FPR. Furthermore, the proposed NAPH module can be utilized in dynamic scenarios in an end-to-end manner. Especially for large-scale dynamic networks, NAPH tracks the multi-scale topology and identify anomalies with high scalability. We also developed

TABLE II
ACRONYMS TABLE

Acronym	Description
NIDS	Network Intrusion Detection System
UNIDS	Unsupervised Network-based Intrusion Detection System
FPR	False Positive Rate
MSF-IDS	Multi-scale Spatial-Frequency Intrusion Detection System
NAPH	Network-aware Persistent Homology
GCN	Graph Convolution Network
RNN	Recurrent Neural Network
MITM	Man-in-the-Middle (attack)
DoS	Denial of Service (attack)
PH	Persistent Homology
TDA	Topological Data Analysis
FWT	Fast Wavelet Transform
NATA	Network-aware Topology Analysis
PD	Persistent Diagram
SOTA	State-Of-The-Art
MLP	Multi-Layer Perceptions
AUC	Aera under Curve (performance metric)
FE	Frequency Encoder
DW	Daubechies Wavelet
SS	Self-supervised representation augmentor
SC	Sub-space Clustering-based intrusion detector

a NAPH application to support visualization and easy-access analysis, which the listed approaches do not equipped.

III. PRELIMINARIES

In this section, we define network traffic flow and its notations, and then provide Definitions 2 to 5. They progressively introduce the theoretical basis of persistent homology to help readers quickly understand it. They also form the necessary theoretical foundations of the following proposed Definitions 6 to 8. Finally, they are used to prove the consistency of our proposed NAPH in Theorem 1. We also list an acronyms Table II, thus enhancing the readability. The detailed notations used in the paper are available in Appendix A (see supplementary material).

Definition 1 (Network Flow): Given a network topology G , considered as an adjacency matrix, $v_i, v_j \in G$ are device nodes with IP address. $F_{ij}(t) = (v_i, v_j, t, \Delta t, \mathbf{F}_{ij})$ is defined as a traffic flow at time t , from the source IP v_i to the destination IP v_j , with duration Δt and traffic feature $\mathbf{F}_{ij}$.

The MSF-IDS inputs are network topology G and features $\mathbf{F}_{ij}$. The MSF-IDS outputs are traffic flow types, e.g., benign, attacks. Then we provide the following concepts for our topological NAPH algorithm.

Definition 2 (Rips Complex): The Vietoris-Rips complex, referred to as the Rips complex, is constructed from a metric space (X, d) with a specified distance threshold $\epsilon > 0$:

$$R_\epsilon(X) = \{\sigma \subseteq X : \forall u, v \in \sigma, d(u, v) \leq \epsilon\}. \quad (1)$$

The Rips complex consists of subsets of points that meet specific distance criteria, and these subsets are closed under inclusion. If $|\sigma| = k + 1$, the subset σ is called a k -simplex, e.g., points, edges, triangles and tetrahedrons corresponding to 0-, 1-, 2-, and 3-simplices.

Definition 3 (Filtration): Filtration decomposes a topological space into nested subcomplexes. Given a topological space X and increasing distance thresholds $\epsilon_0 < \epsilon_1 < \dots < \epsilon_m$, we construct nested Rips complexes $R_{\epsilon_i}(X)$ such that:

$$R_{\epsilon_0}(X) \subseteq R_{\epsilon_1}(X) \subseteq \dots \subseteq R_{\epsilon_m}(X). \quad (2)$$

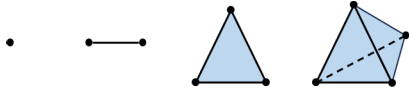


Fig. 3. 0-, 1-, 2-, and 3-simplex from left to right.

Each $R_{\epsilon_i}(X)$ builds topology elements of different scales. However, the above-defined trivial and fundamental topology complexes are not enough to mine subtle topology anomalies in traffic data. We introduce the following definitions:

Definition 4 (Homology Group): Given a Rips complex, homology group $H_k(R_{\epsilon_i}(X))$ is used to quantify its k -dimensional high-order topological information, such as holes or connected components.

The details of $H_k(R_{\epsilon_i}(X))$ are available in Appendix B (see supplementary material). The homology group benefits us from mining the high-order information, e.g., holes, in a given topology. In this way, we have established the high-order topology foundations for the following persistent homology, benefiting the mining of anomaly topology patterns.

Definition 5 (Persistent homology): Given a series of homology groups, generated by nested Rips complexes in Definition 4, persistent homology aims to conduct topological analysis, after employing natural homomorphisms over these groups:

$$H_k(R_{\epsilon_0}(X)) \rightarrow H_k(R_{\epsilon_1}(X)) \rightarrow \cdots \rightarrow H_k(R_{\epsilon_m}(X)). \quad (3)$$

Through these mappings, persistent homology tracks the changes of homology groups $H_k(R_{\epsilon}(X))$ across different scales $\epsilon \in \{\epsilon_0, \dots, \epsilon_m\}$.

More information about PH is available in Appendix J (see supplementary material), which analyses topological information over a series of homology groups in different scales. However, the traditional PH is limited by the filtration with preset ϵ and hard to apply on the real-world evolving network data, e.g., traffic network.

IV. MODEL

The proposed MSF-IDS consists of three critical modules, as shown in Fig. 4. We propose frequency and spatial lenses that highlight spatial-frequency information across multi-scale views. Then we propose spatial-frequency encoders by equipping the lenses. Finally, we propose augmentor and a detector for unsupervised intrusion detection.

A. Multi-Scale Spatial-Frequency Lenses

1) *Wavelet Kernels-Based Frequency Lenses:* Given raw traffic features $\mathbf{F}_{ij}$, frequency lenses serve as a plug-and-play module, preserving multi-scale frequency information. Mathematically, high-frequency information corresponds to features with high eigenvalues. From this perspective, a feasible approach is computing the eigenvector of each eigenvalue. However, this naive method is time-consuming.

To this end, graph filters are used to preserve information more efficiently. Specifically, we propose the following dual graph filters, involving two steps:

Fast Daubechies Wavelet Transformation: First, we introduce the following Daubechies wavelet transformation, which is broadly used as an implementation of fast wavelet transformation, FWT [73]:

$$\mathcal{W}_{\psi}(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi\left(\frac{t-b}{a}\right) dt, \quad (4)$$

where a and b are preset scale hyperparameters. In practice, Eq. 4 is computed using numerical integration techniques, such as Monte Carlo methods. Then the raw traffic features $\mathbf{F}_{ij}$ can be transformed as $\mathcal{W}_{\psi}\mathbf{F}_{ij}$.

The advantages of Daubechies wavelet are two-fold: 1) Daubechies wavelet effectively extracts low and high multi-scale frequency information, the former contains key information of traffic flow, while the latter has the subtle information to distinguish the anomaly traffic. 2) Daubechies wavelet boosts the efficiency of frequency information preserving as a pre-wavelet. it reversibly compresses raw features with 75% compression efficiency theoretically. In practice, we can efficiently conduct Daubechies wavelet transformation by FWT. The specific compression efficiency analysis is available in Appendix E (see supplementary material). We also verify the effectiveness of Daubechies wavelet in the Ablation Study section.

Multi-Scale Beta Wavelet Transformation: Then we use beta kernel-based wavelets to preserve multi-scale frequency information in $\mathcal{W}_{\psi}\mathbf{F}_{ij}$. We have discussed why beta wavelets are capable of preserving multi-scale frequencies, e.g., the high-frequency attack information in Appendix C (see supplementary material). Specifically, we introduce a probability density function of Beta distribution $\beta_{p,q}(w)$:

$$\beta_{p,q}(w) = \begin{cases} \frac{1}{B(p+1, q+1)} w^p (1-w)^q & \text{if } w \in [0, 1] \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

where $p, q \in \mathbb{R}^+$, $B(p+1, q+1) = \frac{p!q!}{(p+q+1)!}$ is a constant, and $w \in [0, 1]$ covers the complete spectral range of traffic features.

We add a restrictions $p, q \in \mathbb{N}^+$ to ensure $\beta^*(p, q)$ is a polynomial, which is same as the previous work [62]. Then the Beta wavelet transform $\mathcal{W}_{p,q}$ can be written as:

$$\mathcal{W}_{p,q} = \frac{(\mathbf{L})^p (\mathbf{I} - \mathbf{L})^q}{B(p+1, q+1)}, \quad (6)$$

where $\mathbf{L}$ is the Laplacian matrix of traffic features. Let $p+q = C$ be a constant and Beta wavelet transform $\mathcal{W}$ is constructed by a group of $C+1$ Beta wavelets with the same order:

$$\mathcal{W} = (\mathcal{W}_{0,C}, \mathcal{W}_{1,C-1}, \dots, \mathcal{W}_{C,0}). \quad (7)$$

Then the raw traffic feature are transformed into different frequencies as follows:

$$\hat{\mathbf{F}}_{ij} = \mathcal{W}(\mathcal{W}_{\psi}\mathbf{F}_{ij}). \quad (8)$$

The frequency lenses effectively preserve the subtle high-frequency information, which is filtered out by traditional GCN-based schemes. The pseudo-code of multi-scale frequency encoders is available in Appendix F (see supplementary material).

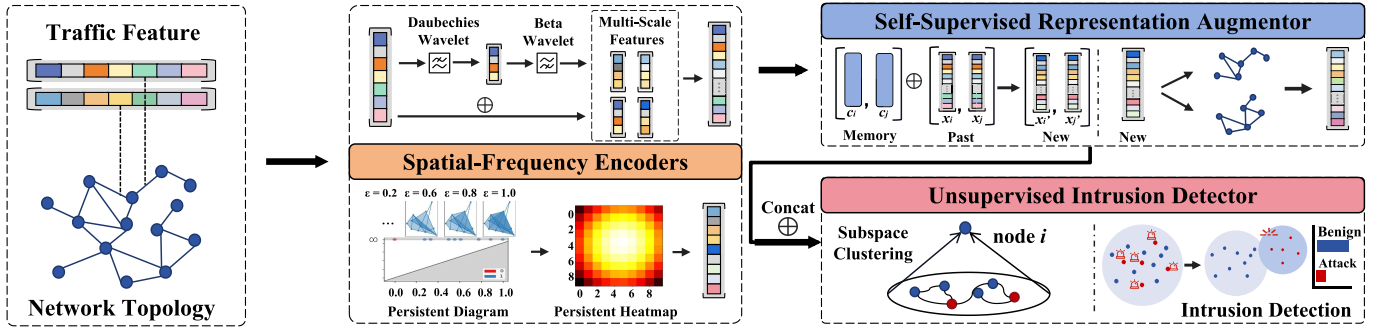


Fig. 4. The architecture of the proposed MSF-IDS. MSF consists of three key components. First, the traffic features and traffic network are fed into the Spatial-Frequency Encoders. The traffic features are handled by the frequency encoder (top) to preserve multi-scale feature information. The traffic network is fed into the spatial encoder (bottom) to track and mine multi-scale topology information. Then the multi-scale features are refined by the Representation Augmentor. Finally, we concatenate the refined feature and topology representations and feed them into the Intrusion Detector to identify suspect traffic.

The frequency lenses help UNIDS effectively preserve subtle high-frequency information, which is filtered out by traditional GCN-based schemes. The pseudo-code of the multi-scale frequency encoders is given in Appendix F (see supplementary material). We verify this benefit empirically in Discussion RQ1.

2) *Persistent Homology-Based Spatial Lenses*: In this section, we formally propose spatial encoders NAPH to track multi-scale topology information in an evolving traffic network. We first propose a specialized filtration as follows:

Definition 6 (Time Filtration): A topological space X is decomposed into a series of time-based nested Rips complexes, defined as $R_{\epsilon(t)}(X)$ for time points $t_0, t_1, \dots, t_m$:

$$R_{\epsilon(t_0)}(X) \subseteq R_{\epsilon(t_1)}(X) \subseteq \dots \subseteq R_{\epsilon(t_m)}(X). \quad (9)$$

At each time point t_i , $R_{\epsilon(t_i)}(X)$ includes all pairs of points (x_i, x_j) such that $d(x_i, x_j) \leq \epsilon(t_i)$, where d represents the distance between points x_i and x_j . This construction allows us to observe and analyze the changes in the complexes over time and varying distance thresholds.

The proposed time filtration enables tracking multi-scale topology in an evolving time stream using $\epsilon(t_i)$, rather than a predefined ϵ . Equipped with time filtration, we propose the following network-aware topology analysis (NATA), enabling tracking multi-scale topology over dynamic traffic data.

Definition 7 (NATA): Given a node in traffic network $x \in G$, where each node $x \in G$ has a timestamp $T(x)$. The node x and its connections form a sub-network $\mathcal{G}(x, t)$ indexed by time t . By introducing time filtration, we generate a series of nested topological complexes $\{R_t\}_{t \in \mathbb{R}}$, where $t_0 < t_1 < \dots < t_m$:

$$R_{t_0}(\mathcal{G}(x, t)) \subseteq R_{t_1}(\mathcal{G}(x, t)) \subseteq \dots \subseteq R_{t_m}(\mathcal{G}(x, t)). \quad (10)$$

Specifically, Eq. 10 specifies the topological space X in Eq. 9 to dynamic network topological space $\mathcal{G}(x, t)$, i.e., dynamic network traffic. Each complex R_{t_i} contains rich topological information of evolving traffic flow within a given time window. Finally, Theorem 1 proves the consistency between NATA and PH theory, enabling multi-scale topology tracking over dynamic traffic networks.

Theorem 1 (Equivalence): The constructed nested topological complexes in NATA (Def. 7) are topologically equivalent

to the nested Rips complexes constructed by filtration in Def. 3.

We rigorously prove the equivalence in Appendix D (see supplementary material) from the perspective of graph isomorphism. Equipped with Theorem 1, we bridge the gap between NATA and persistent homology, terming it network-aware persistent homology (NAPH). The input of NAPH is solely topological information, i.e., network topology. In this way, NAPH tracks multi-scale topology patterns over evolving traffic networks, thereby **addressing Challenge 1**. NAPH involves three steps and its pseudo-code is given in Appendix G (see supplementary material).

We provide three steps to incorporate NAPH into a learnable UNIDS task. NAPH is proposed as an end-to-end topology learning module, thereby **addressing Challenge 2**. More discussions on scalability are available in Appendix R (see supplementary material).

i) *Constructing Nested Traffic Complexes By Time Filtration*: For a target traffic node x , we construct its multi-scale nested complexes $R_t(\mathcal{G}(x, t))$ by the proposed time filtration. $R_t(\mathcal{G}(x, t))$ contains topological information over dynamic network data, i.e., constructing the network complexes using time information t . Note that our NAPH and learnable representations are established on the target traffic node and its sub-network, ensuring the scalability of NAPH.

ii) *Generating Traffic Topological Summaries via NAPH*: Equipped with the equivalence Theorem 1, we employ persistent homology [49] on the traffic nested complexes $R_t(\mathcal{G}(x, t))$ in step 1. PH generates the topological summary of each nested complex. The topological summary can be considered as a profile with evolving multi-scale topological information. Specifically, we generate the barcode $\text{bar}(x, t)$ by analyzing the high-order topological birth-death relationships of traffic complexes. Then the persistent diagrams $PD(x, t)$ with highly refined multi-scale topological information can be generated from barcode $\text{bar}(x, t)$.

iii) *Generating Distinguishable Persistent Heatmap*: By comparing the differences between diagrams, the persistent diagram $PD(x, t)$ with dynamic information is distinguishable for the topology anomalies. However, the diagram $PD(x)$ is not learnable for end-to-end intrusion detection, which means

$PD(x, t)$ is non-scalable. To enhance scalability, we develop a persistent heatmap tool to bridge the gap between persistent diagrams and learnable end-to-end intrusion detection. Specifically, we have the following definition:

Definition 8 (Persistent Heatmap): Given dynamic network data, the persistent heatmap extracts time-aligned statistical information from the persistence diagram, visualizing the multi-scale topology information in color intensity.

$$\text{Heatmap}(x) = f_H(PD(x, t)), \quad (11)$$

where f_H is a transformation and the pseudo-code of f_H is available in the Appendix H (see supplementary material). The persistent heatmap benefits the following spatial encoder to encode distinguishable topology patterns into learnable representations, achieving visualisable, learnable, and differentiable for the multi-scale topological information in dynamic network data.

We also incorporate the above three steps into an executable application, benefiting user personal topological analysis and visualization. NAPH can also seamlessly incorporate into your own end-to-end deep learning model. The details of our NAPH app are available in Section VI and Appendix X (see supplementary material).

B. Multi-Scale Spatial-Frequency Encoders

Equipped with multi-scale spatial-frequency lenses, we formally present spatial-frequency encoders for end-to-end intrusion detection. First, we propose the following frequency encoder for raw traffic features $\mathbf{F}_{ij}$:

$$\mathcal{F}_{ij} = \mathcal{E}_f(\mathbf{F}_{ij}) = \text{CONCAT}(\mathcal{W}(\mathcal{W}_\psi \mathbf{F}_{ij}), \mathbf{F}_{ij}). \quad (12)$$

$\mathcal{F}_{ij}$ carries the multi-scale frequency information in the raw feature $\mathbf{F}_{ij}$. It will be fed into the following representation augmentor for refinement. Furthermore, we propose the following spatial encoders:

$$\mathcal{H}_x = \mathcal{E}_s(\mathcal{G}(x)) = \text{FLAT}(\text{Heatmap}(x)), \quad (13)$$

where $\mathcal{G}(x)$ is a sub-network of node x , $\mathcal{E}_s$ is a spatial encoder. Specifically, a simple yet effective flatten function FLAT as $\mathcal{E}_s$ can be used to transform the two-dimensional image $\text{Heatmap}(x)$ into learnable representation $\mathcal{H}_x \in \mathcal{H}, x \in G$. More details are available in Appendix Q (see supplementary material).

C. Self-Supervised Representation Augmentor

We propose an augmentor to refine the generated multi-scale frequency information. The rationale is two-fold: 1) contrastive rules automatically generate negative samples. We refine the traffic features far from the negative samples without using any labels or prior knowledge. 2) The proposed augmentor simulates real-world noise and disturbances. Through self-supervised learning, representation refinement and generation will be more robust and reliable for UNIDS. The Augmentor involves three steps:

1) *Generating Dynamic Node Representations:* We generate dynamic node representation from evolving traffic edge representations following [41]. Specifically, as for a multi-scale traffic flow $\mathcal{F}_{ij}$, we generate the representation of the node i at t time as $\mathcal{F}_i(t) = \mathcal{F}_i(t^-) + \mathbf{m}_i(t)$, where $\mathcal{F}_i(t^-)$ is the historical representation of node i with initial value $\mathcal{F}_i(0)$ is $\mathbf{0}$. $\mathbf{m}_i(t)$ is the memory of node i at t time. Now, we formally denote $\mathcal{F}$ as the updated dynamic node representations, where $\mathcal{F}_i \in \mathcal{F}, i \in G$. The detailed updating rules are available in Appendix I (see supplementary material).

2) *Augmenting Representations By Dual-View graphs:* Then we generate two graph views G_1 and G_2 , using the corruption function [74]. The function randomly removes a small portion of edges from G and also randomly masks a fraction of dimensions with zeros in node features $\mathcal{F}_i$. The augmented representations of each graph view as:

$$\mathbf{Z} = \text{ReLU}(\hat{G}\text{ReLU}(\hat{G}\mathcal{F}W^{(0)})W^{(1)}), \quad (14)$$

where ReLU is an activate function, $W^{(0)}, W^{(1)} \in W_R$ are shared parameter weights between two graph views, $\hat{G}$ is normalized laplace matrix of G . We simulate disturbance and noise in real-world traffic networks by augmenting feature representations via corrupted dual-view graphs.

3) *Refining Representations By Self-Supervised Learning:* Finally, we refine the node representations $\mathcal{F}$ by proposing a self-supervised loss. The refinement also makes traffic features more distinguishable, using an agreement constraint between dual-view graphs. Specifically, we sample the contrastive samples from augmentations and minimize $\mathcal{L}_R$ as follows:

$$\min_{W_R} \mathcal{L}_R = \sum_{i \in G} \left(-\mathcal{S}_1 + \log \left(\sum_{j \in G-i} [\exp(\mathcal{S}_2) + \exp(\mathcal{S}_3)] \right) \right), \quad (15)$$

where $\mathcal{S}_1 = \frac{\text{sim}(\mathbf{z}_{1i}, \mathbf{z}_{2i})}{\tau}$, $\mathcal{S}_2 = \frac{\text{sim}(\mathbf{z}_{1i}, \mathbf{z}_{1j})}{\tau}$, $\mathcal{S}_3 = \frac{\text{sim}(\mathbf{z}_{1i}, \mathbf{z}_{2j})}{\tau}$, $\mathbf{z}_{1i}$ and $\mathbf{z}_{2i}$ denote embeddings node i on $\mathbf{Z}_1$ and $\mathbf{Z}_2$ respectively, G_{-i} is the node set excluding node i , $\text{sim}(\cdot)$ is the cosine similarity between the two embeddings, and τ is a temperature parameter. The features can be strengthened to counteract the noise by the refinement, without using any labels or prior knowledge.

D. Unsupervised Intrusion Detector

The refined feature representations $\mathbf{Z}$ can be generated from Eq. 14. We concatenate $\mathbf{Z}$ and the topology representations $\mathcal{H}$ as the final representations $\mathcal{Z}$. Then we propose a subspace clustering scheme to identify the attack or suspicious traffic in an end-to-end manner. The detector first derives each traffic node's similarity matrix Q and then outputs the type of each traffic flow, e.g., benign, attack.

$$\min_Q \|\mathcal{Z} - Q\mathcal{Z}\|_F^2 + \lambda_1 \|Q\|_F^2 \quad \text{s.t.} \quad \text{diag}(Q) = \mathbf{0}, \quad (16)$$

where $\|\cdot\|_F$ is a matrix F-norm. Then we generate the traffic membership $\mathbf{C} = \text{Softmax}(\text{MLP}(\mathcal{Z}))$ by matching the similarity matrix Q , $\min_{W_M} \mathcal{L}_M = \|\mathbf{C}\mathbf{C}^T - S\|_F^2$. We have the following intrusion detection loss $\mathcal{L}_{in}$:

$$\min_{W_R, W_M, \mathbf{C}} \mathcal{L}_{in} = \mathcal{L}_R + \mathcal{L}_M. \quad (17)$$

Finally, well-trained traffic memberships $\mathbf{C}$ can be utilized for unsupervised intrusion detection.

TABLE III

COMPARISONS OF BINARY CLASSIFICATION ON FIVE PUBLIC DATASETS. THE RESULTS ARE AVERAGE SCORES OF THREE-TIME REPETITIONS

Methods	CIC-TON-IoT		CIC-BoT-IoT		NF-UQ-NIDS		NF-UNSW-NB15-v2		NF-BOT-IOT-v2	
	F1-score	AUC	F1-score	AUC	F1-score	AUC	F1-score	AUC	F1-score	AUC
AGC [18]	48.42±0.54	52.85±0.64	48.62±0.64	65.52±4.58	45.62±1.00	78.00±0.13	55.83±1.33	58.60±1.09	48.45±0.83	57.96±0.98
DSVD [19]	70.03±0.88	63.22±0.65	43.27±0.33	38.93±0.46	44.23±1.12	60.38±0.74	58.16±0.94	56.62±0.40	42.56±0.81	43.40±0.77
OmniAnomaly [20]	67.49±0.78	64.95±1.16	69.59±1.22	68.29±0.48	60.23±1.36	76.80±1.25	52.60±0.89	60.13±1.47	56.30±0.87	56.19±1.77
SOM-DAGMM [21]	78.13±1.24	69.41±2.13	56.01±0.89	38.83±1.33	<u>62.58±1.42</u>	57.69±0.89	<u>58.92±0.63</u>	50.33±1.04	50.59±2.21	28.15±1.56
EULER [33]	85.83±0.11	91.72±0.05	54.63±0.15	70.22±0.11	53.82±0.12	51.34±0.17	54.38±1.13	50.12±0.76	<u>78.59±1.32</u>	60.18±0.46
APPNP [63]	12.35±0.61	45.30±0.73	14.13±0.51	51.20±0.60	5.57±0.79	44.50±0.79	12.68±0.61	45.90±0.80	15.43±0.73	46.10±0.63
ClenshawGCN [64]	44.22±0.64	72.91±0.18	55.76±0.63	85.59±0.25	18.39±0.11	57.73±1.58	38.32±0.78	<u>75.65±0.34</u>	54.72±0.65	52.94±0.95
AGNN [40]	85.10±0.11	88.57±0.15	80.06±0.84	86.20±1.07	44.61±1.08	42.81±0.94	52.85±1.43	62.03±1.07	56.65±1.32	<u>67.64±2.91</u>
GmapAD [37]	61.17±1.87	53.23±1.48	47.77±1.87	51.67±1.61	48.03±0.58	48.92±0.65	45.81±1.06	52.68±0.80	51.32±0.53	52.27±0.84
MSF-IDS (Ours)	90.55±1.02	91.82±2.68	88.07±0.59	87.93±0.87	65.36±1.90	<u>63.68±3.22</u>	60.76±1.27	76.72±2.63	88.85±2.47	86.15±6.32

V. EXPERIMENTS

A. Experimental Settings

1) *Datasets*:² We performed experiments using 5 extensively utilized datasets that capture significant network traffic within the Internet of Things (IoT). **CIC-ToN-IoT** [39] is derived from the ToN-IoT dataset [75] using the CICFlowMeter tool [76]. The dataset includes 5,351,760 flows, with 53.00% classified as attack samples and 47.00% as benign samples. **CIC-BoT-IoT** [77] is derived from the existing BoT-IoT dataset using the CICFlowMeter tool [78]. The dataset contains 6,714,300 traffic flows, with 98.82% classified as attack samples and 1.18% as benign samples. **NF-UQ-NIDS** [79] is a comprehensive dataset that merges multiple smaller datasets to form a larger, universal NIDS dataset. It includes 11,994,893 records, with 76.77% benign flows and 23.23% attack flows. **NF-UNSW-NB15-v2** [80] is NetFlow-based, originating from the UNSW-NB15 dataset [81]. The dataset contains 2,390,275 traffic flows, with 3.98% identified as attack samples and 96.02% as benign samples. **NF-BoT-IoT-v2** [80] is an IoT NetFlow-based dataset that extends the NF-BoT-IoT dataset [79]. The dataset comprises a total of 37,763,497 data flows, of which 99.64% are attack samples and 0.36% are benign.

2) *Configurations*: All experiments and performance reports are conducted on a machine with Intel Xeon Gold 6330@ 2.00GHz, RTX A40 GPU, and A100 GPU, and on the above 5 datasets with proper data sampling. We set the frequency scale parameter C in the beta wavelet to 4. In NAPH, we predefine the hops of neighbor nodes to 2 with the maximum neighbor node sample number 1024. We use the Adam optimizer with different learning rates for each component and weight decay being $1e^{-5}$. We train all the models within 100 epochs.

3) *Baselines*: We select 5 GCN based-models as baselines, including 2 spatial domain GCN models (i.e., EULER [33], AGNN [40]), 2 spectral GCN model (i.e., ClenshawGCN [64], APPNP [63]), and 1 graph anomaly-based GNN model (i.e., GmapAD [37]). Additionally, we choose 2 anomaly-based baselines (i.e., AGC [18], DSVD [19]), 2 deep learning-based baselines (i.e., OmniAnomaly [20], SOM-DAGMM [21]) to compare with the proposed MSF-IDS.

²The details of datasets, baselines, and implementations are available in Appendix M and Appendix W (see supplementary material).

Metrics: We follow the previous work [82] to evaluate all baselines' performances by two commonly used metrics in intrusion detection, F1-score and ROC-AUC score (AUC).

B. Main Results

1) *Comparisons of Binary Classification*: The proposed MSF-IDS outperforms 9 advanced baselines on 5 benchmarks for the binary classification task. Binary classification identifies whether a traffic flow is benign or an attack, which is conducted on the datasets after proper data sampling. For each comparison method, we run our experiment 3 times and report the mean and variance values. The reproducible results in Table III show that our MSF-IDS consistently performs the best among 9 baselines over the five benchmarks, showing the superiority of our method in unsupervised intrusion detection. Specifically, compared to EULER, the previous state-of-the-art GCN-based UNIDS approach, MSF-IDS achieves 5.50% higher F1-score over the CIC-ToN-IoT dataset. MSF-IDS also outperforms SOM-DAGMM, the previous state-of-the-art deep learning-based method on F1-score by 1.84 points and on AUC by 26.39 points over the NF-UNSW-NB15-v2 dataset, a challenging benchmark for UNIDS. We attribute the gains stem from 1) multi-scale spatial-frequency encoders that differentiate anomaly patterns, 2) augmentor that highlights attack-specific information. To address data imbalance issues, some strategies have been adopted, e.g., re-weight [83] and oversampling [84]. We also observed that the AUC in NF-UQ-NIDS, and F1-score in NF-UNSW-NB15-v2 remain below the supervised ceiling. We leave further improvement to future work.

2) *Comparisons of Multi-Classification*: The proposed MSF-IDS achieves state-of-the-art performance with lower FPR among three selective baselines on the multi-classification task. Multi-classification is a more challenging task to distinguish which type of attacks the suspicious traffic. After appropriate sampling to mitigate imbalance, we compare the proposed MSF-IDS with AGNN, APPNP, and AGC, as they are representative and have been widely used in previous studies. Fig. 5 shows that the proposed MSF-IDS consistently achieves the best results among three baselines on five datasets. Our MSF-IDS also achieves relatively low FPR values across (a) to (d) four datasets. Specifically, MSF-IDS yields the highest F1-scores with 18.85% to 88.30%

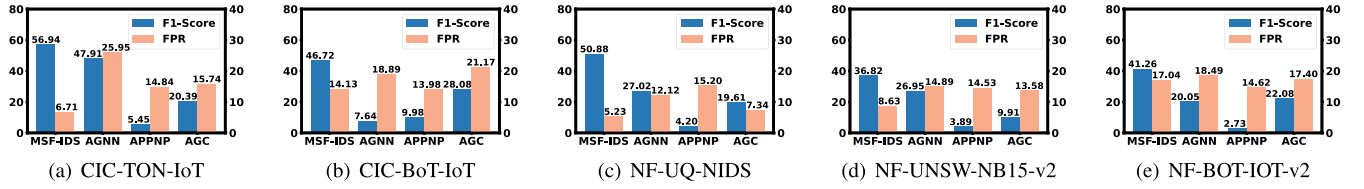


Fig. 5. Comparisons of multi-classification between MSF-IDS and the selective three baselines on five benchmarks.

performance gains over five benchmarks, e.g., outperforms the previous SOTA approach AGNN on CIC-TON-IoT by up to 9.03 points. The performances benefit from multi-scale spatial-frequency designs, thus differentiating anomaly patterns and highlighting attack-specific information. Achieving state-of-the-art F1-score performances, the proposed MSF-IDS also exhibits the lowest FPR across three benchmarks, with reductions ranging from 28.75% to 54.78%. For example, on the CIC-TON-IoT dataset, MSF-IDS achieves an FPR 8.13 points lower than APPNP, reaching as low as 6.71%. The empirical studies in multi-classifications also demonstrate the effectiveness of MSF-IDS in mitigating high FPR. As for the CIC-BoT-IoT dataset, MSF-IDS has a slightly higher FPR, i.e., only 0.15 points than APPNP. However, the F1-score performance of the proposed MSF-IDS is significantly higher than APPNP by 368.14%. Finally, in the NF-BOT-IOT-v2 dataset, the FPR values across various baselines, including the proposed MSF-IDS, remain relatively high, ranging from 14.62% to 18.49%. This highlights the need for further reducing in FPR. Even though, the proposed MSF-IDS still achieves the highest F1-score among all baselines, with an improvement of more than 86.87% compared to other methods.

C. Online Unsupervised Intrusion Detection

We deploy MSF-IDS online for more efficient unsupervised intrusion detection at lower computational time cost. In practice, new traffic samples emerge with time evolving, and frequent retraining is time-consuming and impractical. To this end, a feasible way is to perform incremental online detection. Specifically, for a well-trained UNIDS, we conduct incremental detection for a short period, e.g., one day, and for a long period, e.g., one week, we re-train the UNIDS.

Task setting: We extract 20% traffic in the whole dataset as real-time traffic for online scenario. Then the other 80% traffic is used to train a UNIDS. Finally, the extracted 20% traffic can be detected in an online way.

Baselines and metrics: We select 4 baselines AGC, SOM-DAGMM, and EULER to evaluate MSF-IDS in online scenario. We use F1-Score and NMI values for metrics, following the previous work [33].

As shown in Table IV, the proposed MSF-IDS consistently achieves the highest F1-score and AUC, and the lowest FPR. This empirical study demonstrates that our spatial-temporal encoders are also effective in online settings. Specifically, our MSF-IDS yields 27.16 and 32.69 points improvements in F1-score and AUC, respectively, compared to the previous SOTA GCN-based UNIDS EULER, while MSF-IDS still has 4.60 points lower than EULER in FPR. We also observe that

 TABLE IV
COMPARISONS OF ONLINE INTRUSION DETECTION
ON THE CIC-TON-IOT DATASET

	AGC	SOM-DAGMM	EULER	MSF-IDS (Ours)
F1-score	31.68±0.67	46.73±1.21	50.20±1.46	77.36±1.10
AUC	32.78±0.84	42.43±1.90	46.22±0.35	78.91±1.31
FPR	30.36±0.34	35.73±1.73	32.37±0.88	27.77±0.85

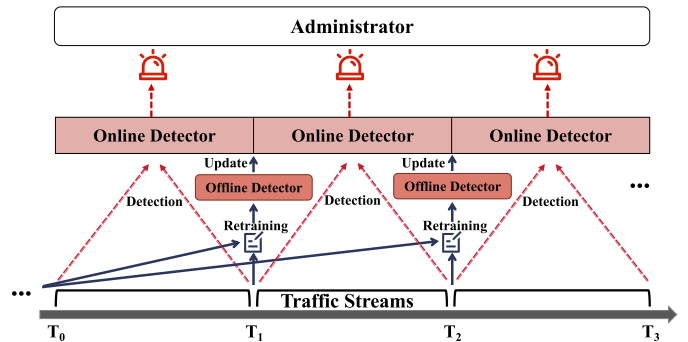


Fig. 6. The practical deployment architecture of the proposed MSF-IDS.

MSF-IDS performs better offline than online. We leave how to close this gap for future work. Nevertheless, the online UNIDS can be an optimal choice, a trade-off, between performance and time cost. We report that 10000 traffic data only requires 10.98s in online detection, while requiring 878.73s in offline detection, involving an expensive time cost in retraining, almost 80 time practical cost. More discussions and future directions are available in Appendix N (see supplementary material).

D. Model Deployment

Furthermore, we present a deployment architecture to illustrate MSF-IDS's future applicability, as shown in Fig. 6. We demonstrate how our model work in real-world traffic streams. Specifically, we adopt an offline-online deployment strategy. In a predefined time window such as T_1T_0 , the online intrusion detector runs for high-efficiency detection. At the end of the time window such as T_1 , the offline detector incrementally retrains and updates the online detector. This strategy effectively balances efficiency and performance without frequent retraining. The online detector can also be empowered by the period update. In real-word scenarios, the time window T_1T_0 can be one day or one week, depending on demands. This hybrid online-offline approach significantly improves detection performance and efficiency. Future challenges include continual updates and adaptive update periods for online detection.

TABLE V

COMPUTATIONAL COMPLEXITY COMPARISONS WITH THE STATE-OF-THE-ART EULER UNDER DIFFERENT TRAFFIC SCALES

Traffic Numbers	EULER	MSF-IDS	FE	NAPH
10,000	41.84s	11.18s	7.49s	0.78s
100,000	422.94s	110.63s	75.20s	7.82s

E. Time Cost Analyses

We provide time cost analyses for MSF-IDS and its critical modules to showcase practical computational superiority. First, we compare the time cost of MSF-IDS and EULER under different traffic scales. As shown in Table V, MSF-IDS is more efficient than the previous SOTA UNIDS approach EULER under 10,000 and 100,000 traffic detection. **One single traffic detection cost within 0.11ms.** Especially for multi-scale spatial mining, our proposed **NAPH framework achieves 10,000 traffic node computation within 1s (0.78s).**

Furthermore, we conduct time cost analyses for the critical modules of the online model. The specific computational cost of critical modules are divided into two parts:

- **Frequency encoder (FE):** FE runs 67.48% of the total time. Fast wavelet transformations (FWT) are proven to handle large-scale data in industry areas efficiently. Using multiple FWTs for FEs to parallelly handle traffic data is feasible and prospective for further acceleration.
- **Spatial encoder (NAPH):** NAPH runs 7.03% of the total time, lying in, 1) limited neighbor nodes sampling strategy (in Configuration), supporting parallel computation of millions of nodes. 2) we use system-inherent operators to achieve computation, e.g., FLAT operator in heatmap, reducing additional complexity.

More importantly, both the FE and NAPH are neural-parametric-free and will not involve computational-intensive backpropagation. We can parallelly detect with a preset batch size, e.g., 10000, thus accelerating the inference speed.

F. Ablation Study

The critical modules such as spatial-frequency encoders, self-supervised representation augmentor, effectively contribute to performances and reduce FPR in UNIDS task. Specifically, we conduct an ablation study on the CIC-ToN-IoT dataset with three repetitions to evaluate the effectiveness of each component. We remove our multi-scale frequency encoders, the proposed Daubechies wavelet, and multi-scale spatial encoders, and denote them as “w/o FE”, “w/o DW”, and “w/o NAPH”, respectively. We use “w/o SS” and “w/o SC” to refer to the model that removes the self-supervised representation augmentor and the subspace clustering-based intrusion detector, respectively.

Table VI reports the comparison results. Specifically, removing the self-supervised augmentor module leads to the most significant performance degradation, e.g., a 14.48 point decrease in F1-score, indicating the necessity for highlighting attack-specific information with the self-supervised term. Our multi-scale spatial-frequency encoder, including the proposed

TABLE VI

ABLATION STUDY OF MSF-IDS ON THE CIC-ToN-IoT DATASET WITH F1-SCORE (%), AUC (%), AND FPR (%), RECALL (%)

Variants	F1-score	AUC	FPR
w/o FE	85.91± 2.92	86.13± 2.66	26.78± 1.46
w/o DW	84.64± 2.43	86.85± 2.55	18.25± 1.89
w/o NAPH	87.40± 2.99	87.17± 3.04	25.45± 11.29
w/o SS	76.66± 2.95	76.36± 2.20	22.38± 2.63
w/o SC	84.97± 1.32	84.45± 1.29	18.45± 9.33
MSF-IDS (ours)	91.14± 0.81	91.82± 1.00	9.94± 0.83

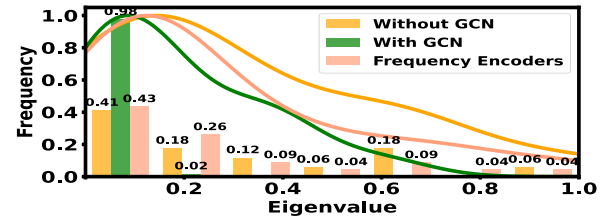


Fig. 7. The spectra distribution between different schemes.

Daubechies wavelet and self-expressive detector are also non-trivial to push the F1-score to 92.22%, with 3.74% - 5.23% performance gains. More importantly, MSF-IDS achieves the lowest FPR among different variants. In contrast, without frequency or spatial encoders leads to the highest FPR, verifying the effectiveness of MSF-IDS in mitigating high FPR.

G. Discussion

RQ1: How do our wavelet-based frequency encoders help distinguish the anomalous traffic? First, the quantitative results are available in the ablation study. The F1-score and the FPR of w/o FE and MSF-IDS significantly demonstrate the effectiveness of frequency encoders. Furthermore, we conduct a visualized study to demonstrate whether our frequency encoders help distinguish the anomaly patterns. Specifically, we depict the frequency distributions in different schemes, i.e., the raw features, features handled with GCN, and features handled with our frequency encoders. As shown in Fig 7, the distribution of frequency encoders has mitigated the ‘left-shift’ phenomenon, preserving more high-frequency information, compared to the distribution handled with GCN. Future directions on spatial-frequency attack and defense are available in Appendix S (see supplementary material).

RQ2: How does our NAPH highlight the anomalous traffic topology? To answer this question, we analyze a MITM case using our NAPH application. We first run NAPH on a benign topology, top box of Fig. 7, and then on the MITM attack topology, bottom box of Fig. 8, with a red attack node UE6. The visualized results of heatmaps are highly distinguishable in (d) and (h). It shows that our spatial encoders effectively identify the anomaly pattern of MITM, a tough case when using the existing topology learning approaches [85], [86]. Most existing approaches are built on the cluster assumption [87], leading to two topological patterns in (a) and (b) undistinguishable. To further reveal the underlying reason, we analyze the barcodes in (b) and (f). The differences

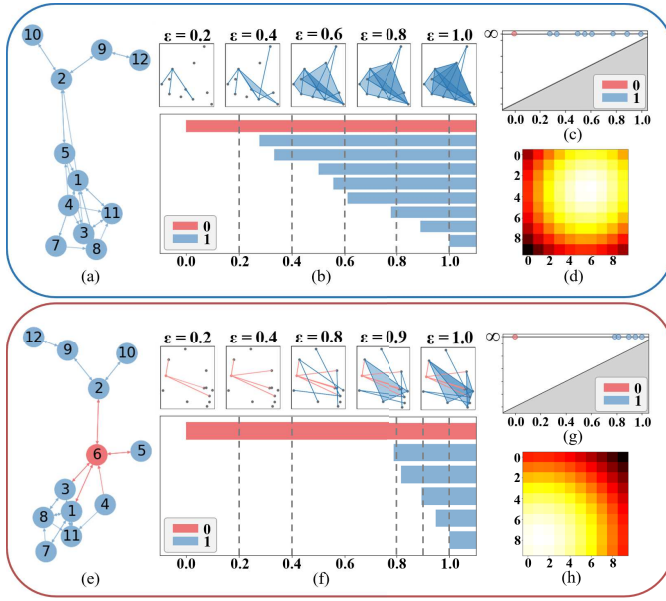


Fig. 8. NAPH analysis for benign and attack traffic topology, respectively. (a)(e) traffic networks. (b)(f) topological barcodes. (c)(d) persistent diagrams correspond to barcodes. (d)(h) heatmaps generated via persistent diagrams.

TABLE VII
COMPARISONS OF FPR ON FIVE DATASETS

	CIC-TON-IoT	CIC-BoT-IoT	NF-UQ-NIDS	NF-UNSW-NB15-v2	NF-BOT-IOT-v2
DSVD	37.21	39.00	76.31	41.52	32.87
SOM-DAGMM	15.88	67.86	44.48	49.53	84.85
OmniAnomaly	52.35	35.01	68.84	56.71	60.77
GmapAD	61.95	36.34	54.17	42.67	45.06
MSF-IDS (Ours)	10.90 (↓ 31.36%)	24.22 (↓ 30.82%)	35.19 (↓ 20.89%)	27.50 (↓ 33.77%)	2.42 (↓ 92.64%)

in barcodes directly determine the distinguishable heatmaps. First, the barcode of the benign one shows longer persistent times in high-order structure, the blue color ones, compared to the anomaly one. It is related to potential stable communication topology pattern construction. These patterns are more persistent and robust than the patterns in MITM attack. As shown in Fig. 8(e) the attack node UE6 blocks the normal communications of other attacked nodes. It is one of the reasons why MITM cannot establish stable patterns.

RQ3: Can the spatial-frequency encoders reduce the false positive rate in binary classification? To answer this question, we conduct FPR comparisons in binary classification on five datasets, as shown in Table VII. The proposed MSF-IDS achieves the lowest FPR among 4 selective baselines, 20.89% - 92.64% decreases, compared to previous approaches. For example, in the CIC-ToN-IoT dataset, our MSF-IDS yields a 31.36% decrease from deep learning-based SOM-DAGMM.

RQ4: How do the augmentor and detector highlight attack-specific information? To answer this question, we visualize the representations with and without the augmentor and detector on the CIC-TON-IoT dataset via t-SNE technology. As shown in Fig. 9(a), representations without the augmentor and detector are indistinguishable and are very close and similar in low-dimension space, i.e., the red round box. It is an important visualization evidence of why without augmentor and detector leads to poor performance as shown in ablation studies. In contrast, the representations are highly

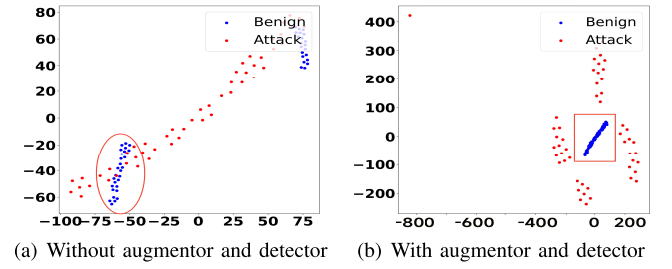


Fig. 9. The t-SNE visualizations of representations on the CIC-TON-IoT dataset without and with the augmentor and detector.

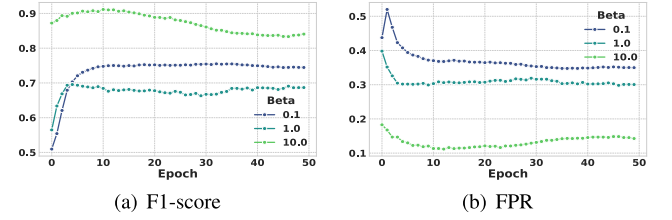


Fig. 10. The sensitivity analysis of hyperparameter Beta.

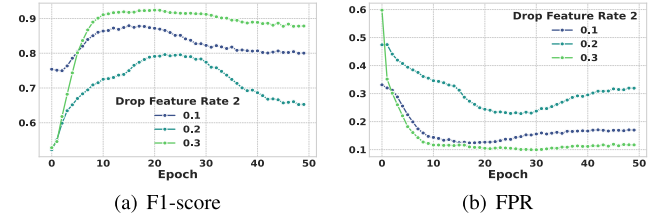


Fig. 11. The sensitivity analysis of hyperparameter drop feature rate.

separated and distinguishable when using the augmentor and detector as shown in Fig. 9(b), i.e., the square box. It significantly verifies the effectiveness of our augmentor and detector in highlighting attack-specific information. We further discuss novel attack detection ability of the proposed MSF-IDS in Appendix T (see supplementary material).

H. Hyperparameter Sensitivity Analyses

We conduct hyperparameter sensitivity analyses on the CIC-BoT-IoT dataset to demonstrate how the critical hyperparameters affect MSF-IDS. We select two critical hyperparameters, Beta, which controls the loss intensity and drop feature 2, which control the feature learning intensity of the representation augmentor. As shown in Fig. 10 and Fig. 11, the proposed MSF-IDS stably converged to the best performance with increasing epoch. Specifically, a larger beta and drop rate 2 perform better, demonstrating the effectiveness of mitigating FPR and boosting performance. More hyperparameter sensitivity analyses are available in Appendix V (see supplementary material).

Finally, we depict performance curves on the CIC-ToN-IoT dataset with different feature space dimensions, 64, 128, and 256, respectively. As shown in Fig. 12, all dimensions settings are stable and converge to the best performance as FPR decreases. More performance curves on various benchmarks in different dimensions are available in Appendix U

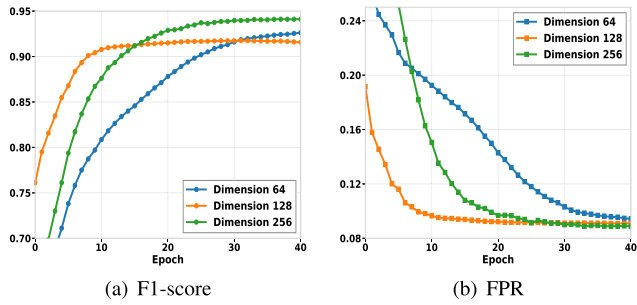


Fig. 12. The F1-score and FPR comparisons on the CIC-ToN-IoT dataset with different feature space dimensions.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Flow ID	Src IP	Src Port	Dst IP	Dst Port	Protocol	Timestamp	Flow Dura	Tot Fwd P	Tot Bwd P	TotLen Fw	TotLen Bw	Fwd Pkt L	Fwd Pkt U	Bwd Pkt L	Bwd Pkt U
2	192.168.1.1	192.168.2.1	62985	192.168.1.1	80	6/04/06/201	24789150	3	3	100	100	100	0	33.3333	57.73503	100
3	192.168.1.1	192.168.2.1	34110	192.168.2.1	80	17/04/06/201	20284432	6	1	0	0	0	0	0	0	0
4	192.168.1.1	192.168.3.1	55130	192.168.2.1	80	6/04/06/201	26449174	3	3	100	100	100	0	33.3333	57.73503	100
5	192.168.1.1	192.168.1.1	18813	192.168.1.1	80	6/04/06/201	13105976	2	1	200	100	100	100	100	0	100
6	192.168.1.1	192.168.2.1	52929	192.168.2.1	80	6/04/06/201	18775087	5	2	400	100	100	0	80	44.72136	100
7	192.168.1.1	192.168.1.1	11551	192.168.2.1	80	17/04/06/201	38594445	9	1	0	0	0	0	0	0	0
8	192.168.1.1	192.168.2.1	40850	192.168.2.1	41269	6/21/05/201	91	0	2	0	0	0	0	0	0	0
9	192.168.1.1	192.168.1.1	36132	192.168.6.1	80	6/04/06/201	14618756	5	2	400	100	100	0	80	44.72136	100
10	192.168.1.1	192.168.1.1	13332	192.168.2.1	80	17/04/06/201	17567581	4	1	0	0	0	0	0	0	0
11	192.168.1.1	192.168.1.1	36882	192.168.2.1	80	6/04/06/201	22435271	6	2	500	100	100	0	83.3333	40.82483	100
12	192.168.1.1	192.168.1.1	14529	192.168.1.1	80	17/04/06/201	3553104	11	1	0	0	0	0	0	0	0
13	192.168.1.1	192.168.1.1	60385	192.168.1.1	80	17/04/06/201	13738110	6	1	0	0	0	0	0	0	0
14	192.168.1.1	192.168.2.1	36102	192.168.1.1	80	6/04/06/201	23397315	3	1	300	100	100	100	100	0	100
15	192.168.1.1	192.168.8.1	50472	192.168.2.1	80	17/04/06/201	25808710	4	1	0	0	0	0	0	0	0
16	192.168.1.1	192.168.1.1	61860	192.168.2.1	80	6/04/06/201	28977431	5	2	400	100	100	0	80	44.72136	100
17	192.168.1.1	192.168.2.1	14008	192.168.2.1	80	17/04/06/201	13519504	10	1	0	0	0	0	0	0	0
18	192.168.1.1	192.168.1.1	35360	192.168.1.1	80	17/04/06/201	21998972	4	1	0	0	0	0	0	0	0
19	192.168.1.1	192.168.7.1	42969	192.168.1.1	80	17/04/06/201	13173022	6	1	0	0	0	0	0	0	0
20	192.168.1.1	192.168.1.1	42643	192.168.1.1	80	6/04/06/201	24482267	6	3	400	100	100	0	66.6667	51.63978	100
21	192.168.1.1	192.168.1.1	26601	192.168.2.1	80	17/04/06/201	19071750	12	1	0	0	0	0	0	0	0
22	192.168.1.1	192.168.2.1	4355	192.168.1.1	80	6/04/06/201	29722842	4	1	400	100	100	100	100	0	100
23	192.168.1.1	192.168.1.1	41453	192.168.1.1	80	6/04/06/201	14019389	6	1	600	100	100	100	100	0	100
24	192.168.1.1	192.168.1.1	51489	192.168.1.1	80	6/04/06/201	23920553	5	2	400	100	100	0	80	44.72136	100
25	192.168.1.1	192.168.1.1	2849	192.168.1.1	80	6/04/06/201	9023446	2	1	200	100	100	100	100	0	100
26	192.168.1.1	192.168.7.1	36302	192.168.7.1	80	17/04/06/201	20933193	6	1	0	0	0	0	0	0	0
27	192.168.1.1	192.168.1.1	10821	192.168.1.1	80	6/04/06/201	18491720	7	1	700	100	100	100	100	0	100
28	192.168.1.1	192.168.4.1	12999	192.168.3.1	80	6/04/06/201	15676198	6	1	600	100	100	100	100	0	100

Fig. 13. The Raw traffic data. NAPH processes them in an end-to-end way.

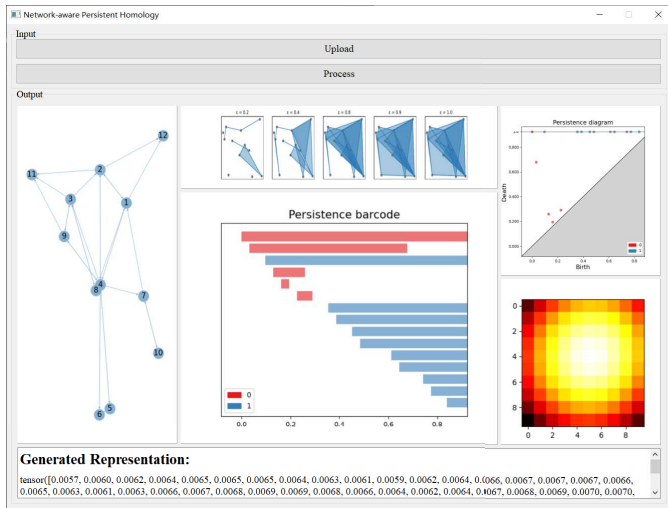


Fig. 14. A topological case of our NAPH, the visualization can be used for topological data analysis, e.g., traffic topology, geological analysis, social, financial network analysis, etc.

(see supplementary material). The empirical results show that sufficient, diverse datasets and multi-scale spatial-frequency preservation schemes effectively mitigate overfitting.

VI. NAPH APPLICATION

To enable visual analysis of multi-scale traffic topology, especially for readers unfamiliar with PH and provided source codes, we develop an executable (.exe) NAPH application. It supports the **real-time multi-scale topology computation** for input dynamic traffic data using NAPH algorithm.

As shown in Fig. 13, a data example we provided, NAPH can automatically extract the network topology information for multi-scale analyses. After processing the data, as shown in Fig. 14, NAPH provides distinguishable heatmaps of multi-scale topology. The representations can be used to end-to-end deep learning, e.g., UNIDS task. Specifically, NAPH displays: input topology, i.e., left side, dynamic topology evolving and persistent barcode, i.e., center, persistent diagram and our persistent heatmap, i.e., right side. Finally, the generated representation is shown at the bottom and can be seamlessly incorporated into end-to-end deep learning. The future practicability of NAPH in various security tasks are available in Appendix O (see supplementary material), including IoT security enhancement, Industrial Control Systems (ICS) Protection, and Zero-Day attack detection, etc.

VII. CONCLUSION

This paper reveals that the high FPR of GCN-based UNIDS stems from a single-scale spatial-frequency learning paradigm. This interesting observation motivates us to propose MSF-IDS, mitigating high FPR by mining multi-scale spatial-frequency clues. Specifically, the proposed MSF-IDS employs dual graph filters as frequency encoders, differentiating high-frequency abnormal patterns. Then a NAPH module is seamlessly integrated into MSF-IDS to track multi-scale topological patterns over dynamic networks. The developed NAPH app also benefits the topological analysis and visualizations of network data, e.g., traffic network. Our MSF-IDS also refines and highlights attack-specific information through the proposed representation augmentor and intrusion detector. Finally, extensive experiments including binary, multi-class, online intrusion detection and discussions demonstrate MSF-IDS's superiority over the traditional GCN-based UNIDS, effectively reducing false positive rates and improving detection performance. NAPH also sheds light on effective multi-scale topology mining over dynamic networks, such as financial, social, and geometric networks. Future work can extend to effective small-sample or few-shot learning in UNIDS.

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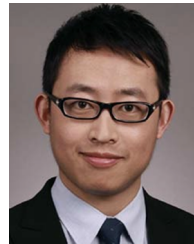
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